What pointing taught us about embodied AI.

FINDING 01

The cost of embodiment.

Making physical pointing accurate at room scale is non-trivial: it requires **custom inverse kinematics, coordinate mapping, laser offset calculation, and servo calibration.** Embodied AI communication carries a real technical cost that screen and voice-based systems simply do not have to pay.

↳ HARDWARE · IK · CALIBRATION

FINDING 02

Form follows interaction.

The shift from a robotic arm to a pan-tilt laser was not only an aesthetic decision — it changed the **quality of the interaction itself:** what it felt like to be pointed at, and what the robot communicated through its physical presence in the room.

↳ FORM · PRESENCE · AFFECT

FINDING 03

A language already known.

When people encountered PointoBot for the first time, the **onboarding cost dropped to zero.** Rather than forcing users to learn a new interface, the AI adopted a gesture humans have known since birth — the same one a child uses before they have words.

↳ LEGIBILITY · ONBOARDING · GESTURE